

Cross-Platform Colon Cancer Proliferation Classification

via Leakage-Free ML: Calibration Benchmark & Custom StabilitySelector Algorithm

ScienceMontgomery 2025 | Ronisnotasianfr | Project ID: CS-XXX

GEO (GSE39582, GSE17538) • TCGA-COAD/READ • CPTAC-COAD • GDSC2 (295 drugs)

Problem & Motivation

Colon cancer is the 2nd leading cause of cancer death worldwide. Ki-67 IHC is the clinical gold standard for proliferation assessment, but it is subjective, single-gene, and cannot predict drug response.

Can machine learning predict proliferation class from transcriptomic data across platforms?

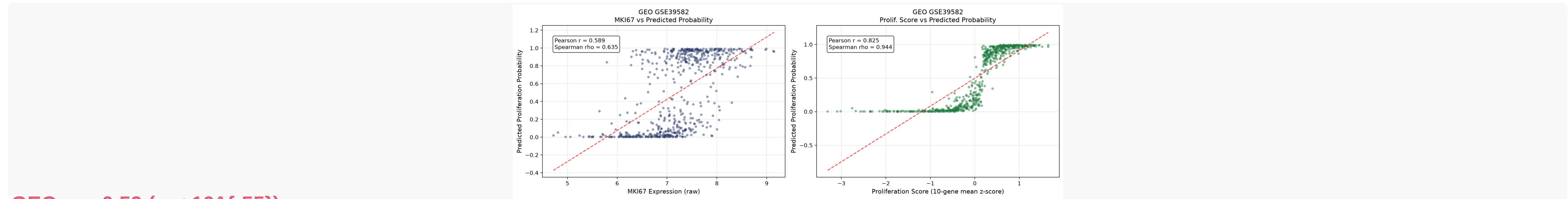
5 calibration methods x 4 model classes

StabilitySelector: custom bootstrap feature selector

Bonferroni-corrected screen of 295 drugs

Cross-platform validation (GEO, TCGA, CPTAC)

Ki-67 Correlation (Leakage-Free)

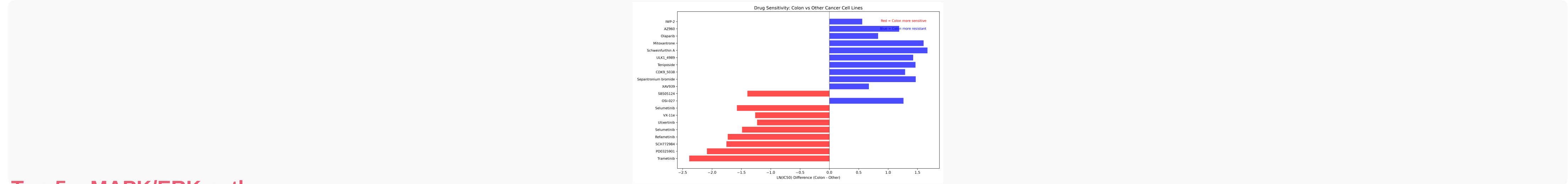


GEO: $r = 0.59$ ($p < 10^{-55}$)

TCGA-COAD: $r = 0.54$ ($p < 10^{-25}$)

Gene held out — model finds proliferation via downstream transcriptional cascades

Drug Sensitivity (GDSC2)



Top 5 = MAPK/ERK pathway

Trametinib (MEK) $p = 1.8e-12$

PD0325901 (MEK) $p = 5.9e-12$

SCH772984 (ERK) $p = 1.1e-10$

Pipeline & Methodology

GEO GSE39582 (585) + GSE17538 (232) -> train
Remove 10 proliferation genes (leakage prevention)
StabilitySelector: bootstrap 100x, keep stable ($\geq 50\%$)
4 models: Logistic Regression, RF, XGBoost, MLP
Nested CV: GridSearchCV + 5-fold StratifiedKFold
TCGA-COAD (329) + CPTAC (105) -> cross-platform validate
Calibration: Platt, Isotonic, QN+Platt, QN Only
StabilitySelector: custom algorithm

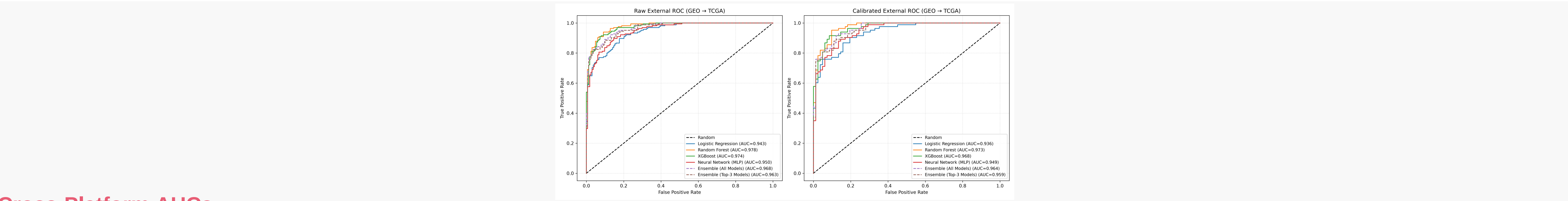
StabilitySelector: custom algorithm

Resamples data B=100 times, computes ANOVA F-scores

Retains features selected in $\geq 50\%$ of resamples

More robust to data perturbations than SelectKBest

External Validation: GEO -> TCGA



Cross-Platform AUCs

Logistic Regression Raw AUC: 0.974 Calibrated: 0.963

Random Forest Raw AUC: 0.962 Calibrated: 0.952

XGBoost Raw AUC: 0.971 Calibrated: 0.964

Neural Network (MLP) Raw AUC: 0.975 Calibrated: 0.967

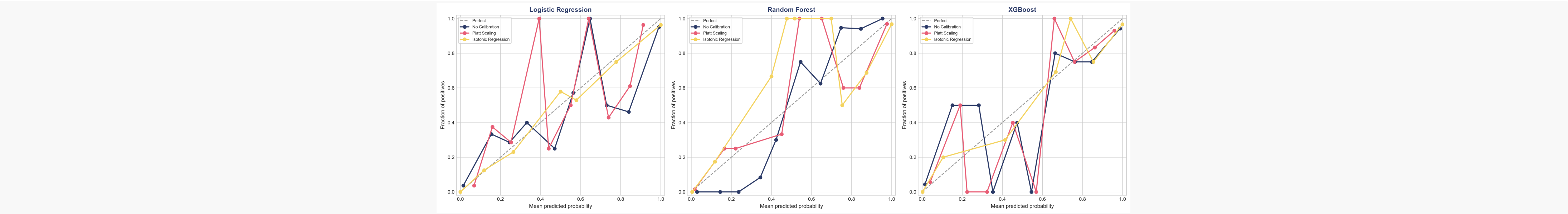
Ensemble (All Models) Raw AUC: 0.976 Calibrated: 0.967

Calibration Benchmark Results

Key Finding: Platt Scaling Significantly Improves ECE

RF Platt: AUC=0.973, ECE=0.043 (CI non-overlapping with No Calibration)

LR QN+Platt: AUC=0.972 (best cross-platform calibration)



Best Configurations

Logistic Regre Isotonic Regress AUC=0.925 ECE=0.028

Neural Network Isotonic Regress AUC=0.935 ECE=0.029

XGBoost Platt Scaling AUC=0.968 ECE=0.038

Random Forest Platt Scaling AUC=0.973 ECE=0.043

XGBoost Isotonic Regress AUC=0.961 ECE=0.044

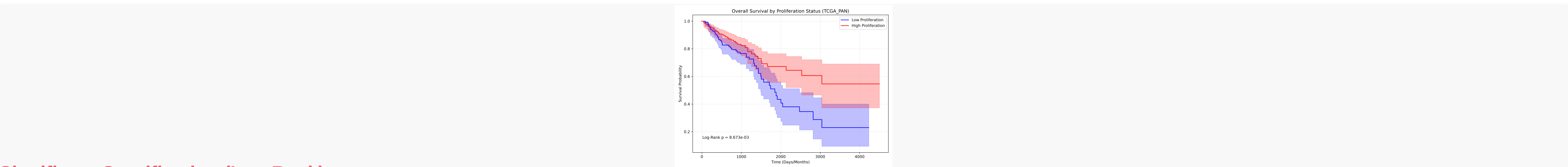
XGBoost No Calibration AUC=0.968 ECE=0.058

Ablation: StabilitySelector vs SelectKBest

Result: StabilitySelector matches or beats SelectKBest

MLP: +0.009 AUC improvement | Others: equivalent

Survival Analysis



Significant Stratification (Log-Rank)

GSE39582 (GEO) $p = 0.037$

TCGA-COAD $p = 0.034$

TCGA PanCancer $p = 0.009$